Arya Koureshi

Senior AI Engineer | Full-Stack Developer

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GitHub • Portfolio

PROFESSIONAL SUMMARY

Senior AI Engineer and Full-Stack Developer with approximately two years of product-development experience at Done Institute, delivering React and Node.js platforms from concept to production. Combines hands-on web engineering with a research background in computer vision, medical imaging, deep learning, and EEG/BCI signal analysis. Experienced across the full lifecycle: data preparation, model development and evaluation, API and UI implementation, production troubleshooting, and iterative product delivery.

~2 years	3 products	47 repos	290+ stars
Done Institute	React + Node.js platforms	Public GitHub portfolio	Across public repositories

TECHNICAL EXPERTISE

AI & Deep Learning: Python, PyTorch, TensorFlow/Keras, scikit-learn, CNN, U-Net, YOLO, ViT, EfficientNet, transfer learning, autoencoders, GANs.

Computer Vision: OpenCV, object detection, segmentation, classification, OCR/Tesseract, real-time video inference, augmentation, denoising, morphology.

Full-Stack Engineering: JavaScript, React, Node.js, REST APIs, responsive UI, authentication flows, dashboards, file-based workflows, HTML/CSS, Git/GitHub.

Biomedical AI & Research: EEG/BCI, CSP/FBCSP, CCA, P300/SSVEP, PSD, STFT, source localization, MATLAB, NumPy, pandas, Jupyter.

PROFESSIONAL EXPERIENCE

Senior AI Engineer & Full-Stack Developer | Done Institute — 2024 – 2026 · ~2 years

Product engineering, applied AI, and end-to-end web delivery

- Designed and delivered three user-facing platforms with React front ends and Node.js back ends, covering product structure, reusable UI, API integration, testing, deployment, and post-launch iteration.
- Built multi-step user workflows for service discovery, request submission, account-based tracking, and operational follow-up; translated business requirements into maintainable product features.
- Applied senior AI engineering experience to model selection, data pipelines, computer-vision prototypes, evaluation, and integration planning for practical product and research use cases.
- Worked across both research-grade code and production-facing systems, bridging experimentation, engineering constraints, usability, performance, SEO, and release readiness.

DoneProject
React · Node.js
Academic project-services platform with structured request and delivery workflows.

DoneGram
React · Node.js
Social-networking platform created for the Done Institute community.

Daramhavato
React · Node.js
Live academic-services platform with online requests, file submission, and status tracking.

AI / BCI Researcher | Ghazizadeh Lab & Sharif Brain Center, Sharif University of Technology — Apr 2023 – Present

- Develop real-time motor-imagery EEG pipelines for BCI and exoskeleton-oriented control, spanning preprocessing, feature extraction, classification, denoising, and visualization.
- Implement and evaluate CSP/FBCSP, CCA, P300/SSVEP, GEVD/DSS, PSD, STFT, source-localization, and neuroscience-modeling workflows in Python and MATLAB.
- Convert research concepts into reproducible notebooks, analyses, reports, and codebases for applied biomedical AI experimentation.

Teaching Assistant — Learning in Brain and Machine | Sharif University of Technology — Sep 2023 – Present

- Support instruction in neural computation, brain-inspired learning, artificial neural networks, and machine learning under Prof. Reza Ebrahimpour.

Teaching Assistant — Project-Based AI Summer Camp | Institute for Research in Fundamental Sciences (IPM) — Jul 2023 – Nov 2023

- Mentored applied projects in PyTorch, TensorFlow/Keras, OpenCV, CNNs, transfer learning, video processing, VAEs/GANs, sequence models, NLP, audio, and deployment fundamentals.

SELECTED AI, COMPUTER VISION & DATA PROJECTS

Brain Tumor Detection & Segmentation 74 stars · 12 forks
Medical-image segmentation pipeline using CNN/U-Net, including annotation conversion, mask generation, preprocessing, augmentation, training preparation, and result visualization.

EEG Signal Processing & Analysis 40 stars · 6 forks
Python workflow for filtering, downsampling, zero-padding, windowing, signal generation, and time-frequency analysis with STFT; part of a broader public EEG/BCI portfolio.

Persian License Plate Recognition YOLO · TensorFlow
End-to-end detection-to-recognition pipeline combining YOLO localization with a TensorFlow/Keras character classifier, dataset preparation, evaluation, confusion matrices, and visual inspection.

Medical Diagnostic Imaging Models VGG · ViT · EfficientNet
Implemented classification workflows for mammography, chest X-ray pneumonia, and skin-lesion diagnosis, covering preprocessing, imbalance handling, training, evaluation, and visualization.

Real-Time Emotion Detection OpenCV · VGG16
Image and real-time video emotion recognition using baseline/deeper CNNs and VGG16 transfer learning, with data preparation, model comparison, and live inference.

Neon Eye-Tracking & IMU Analysis Python · Real-time data
Captures synchronized video and gaze data from Pupil Labs Neon, stores recordings locally, and supports downstream analysis with Python data-science tools.

Motor-Imagery EEG Classification CSP · SVM/LDA/KNN
Two-class mental-task classification using CSP features, MNE-based processing, multiple classifiers, and k-fold cross-validation; complemented by public P300, SSVEP, GEVD/DSS, and denoising projects.

CFD & Experimental Image Analysis 14 stars · Python
Simulation and image-processing workflow for analyzing gas-bubble hydrodynamics in tapered fluidized beds, combining experimental data, computational modeling, and quantitative analysis.

Medical Image Processing Suite OpenCV · scikit-image
Portfolio of medical-image enhancement and denoising methods spanning Fourier analysis, PCA, morphology, NLM, Total Variation, adaptive filters, low-rank techniques, and WNNM.

SolSudo — Image-Based Sudoku Solver 10 stars · OCR/CNN
Practical vision application combining image preprocessing, Tesseract OCR, neural digit prediction, grid extraction, and an automated puzzle-solving pipeline.

EDUCATION

Ph.D. Student | University of Tehran 2024 – Present
Focus: artificial intelligence, machine learning, and biomedical signal/image analysis.

M.Sc., Biomedical Engineering (Bioelectric) | Sharif University of Technology Sep 2022 – Jul 2024
Research: EEG/BCI, motor imagery, computational neuroscience, and brain-machine interfaces.

B.Sc., Biomedical Engineering | University of Tabriz Sep 2018 – Jul 2022
Thesis: implementation of FBCSP for EEG-based motor-imagery BCI.

CERTIFICATIONS

- Deep Learning — Neuromatch Academy, 2023
- Artificial Intelligence & Deep Learning — IPM, 2023
- 5th Sharif Neuroscience Symposium, 2023
- From Single Neuron to Learning in Networks — IPM, 2023
- g.tec Medical Engineering Spring School, 2021
- Brain Mapping Laboratory Program, 2020
- Fundamental Course of MATLAB, 2019

ADDITIONAL EXPERIENCE

Translator, Invigilator & Mathematics Olympiad Teacher
Erdős Institute · Oct 2019 – Present
Supported international mathematics and science competitions, including SIMOC, SASMO, SINGA, AMO, IJMO, DrCT, VANDA, and VTMO.

Member | Iranian Neuroscience Society · 2023 – Present
Selected contributor of proposed Iranian questions for Kangourou Sans Frontières in 2018 and 2020.

GitHub portfolio figures verified July 20, 2026. Links are clickable in the PDF.